

Supplementary material of Predicting small-molecule inhibition of protein complexes

Table 1 Detail of 2P2I Dataset						
Sr. No	PPI Target	Complex Name	Positive	Negative	N/P ratio	Names of chains
1	TNFA/TNFA	1TNF	1	-	-	['A', 'B']
2	TNFR1A/TNFB	1TNR	1	-	-	['B', 'A']
3	HPV-E2/E1	1TUE	1	-	-	['A', 'B']
4	XDM2/P53	1YCQ	11	320	29.1	['A', 'B']
5	INTEGRASE/LEDGF	2B4J	65	267	4.1	['A', 'B']
6	MDM4/P53	3DAB	5	314	62.8	['A', 'B']
7	CREBBP/H4	2RNY	61	324	5.3	['A', 'B']
8	BRD4-1/H4	3UVW	201	1061	5.3	['A', 'B']
9	MDM2/P53	1YCR	51	1220	23.9	['A', 'B']
10	IL-2/IL-2R	1Z92	7	30	4.3	['A', 'B']
11	XIAP-BIR3/SMAC	1NW9	13	144	11.1	['A', 'B']
12	ZIPA/FTSZ	1F47	4	17	4.2	['B', 'A']
13	MENIN/MLL	4GQ6	23	173	7.5	['B', 'A']
14	BCLXL/BAK	1BXL	12	65	5.4	['B', 'A']
15	BCL2/BAX	2XA0	3	1616	538.7	['B', 'A']
16	HRAS/SOS1	1BKD	2	90	45	['R', 'S']
17	KEAP1/NRF2	2FLU	12	508	42.3	['X', 'P']
18	BRD9/H4	4YY6	22	100	4.5	['Z', 'A']
19	BAZ2B/H4	4QC3	104	295	2.8	['C', 'B', 'A']
20	VHL/HIF1A	4AJY	90	279	3.1	['C', 'V', 'H', 'B']
21	WDR5/MLL1	4ESG	30	1160	38.7	['D', 'C', 'A', 'B']
22	CIAP1-BIR3/CASPASE-9	3D9T	28	633	22.6	['C', 'A', 'D', 'B']
23	MKEAP1/MNRF2	3WN7	27	1446	53.6	['L', 'A', 'M', 'B']
24	BRD2-2/H4	2E3K	66	281	4.3	['R', 'B', 'A', 'C', 'Q', 'D']
25	DCN1/UBC12	3TDU	20	70	3.5	['E', 'F', 'A', 'D', 'B', 'C']

Table 2 Amino acid seven classes partition based on their physicochemical characteristics.				
Class	Amino acid	Dipole moment	Volume (Å ³)	Diffused bond creation
1	A, V, G	<1.0	<50	No
2	I, F, L, P	<1.0	>50	No
3	M, S, T, Y	(1.0, 2.0)	>50	No
4	H, N, Q, W	(2.0, 3.0)	>50	No
5	K, Rs	>3.0	>50	No
6	D, E	>3.0(opposite direction)	>50	No
7	C	<1.0	<50	Yes

Table 3 External dataset collect from recent publications			
Sr. No	Inhibited Complex	Inhibitor Name	SMILES
1	7P58	5UI	CC(C)Oc1cccc(c1)c2cccc(n2)n3c(c(cn3)C(=O)O)C(F)(F)F
2	7P5E	5VX	CN(C)C(=O)c1cccc(c1)c2cccc(n2)n3c(c(cn3)C(=O)O)C(F)(F)F
3	7P5F	5QZ	CN(C)C(=O)c1cccc(c1)c2cccc(c2)n3c(c(cn3)C(=O)O)C4CC4
4	7P5I	5X9	Cn1cc(nn1)C2CC2c3c(cnn3c4cccc(c4)c5cccc(c5)C(=O)N(C)C)C(=O)O
5	7P5K	5UZ	CCCC1CCCCN1C(=O)c2cccc(c2F)c3cccc(c3)n4c(c(cn4)C(=O)O)C5CC5
6	7P5N	5RQ	CCCCC1CCCN1C(=O)C2CCCC(C2)c3cccc(c3)n4c(c(cn4)C(=O)O)C5CC5
7	7P5P	5RB	CCCC1CCCCN1C(=O)c2cccc(c2)c3cccc(c3)n4c(c(cn4)C(=O)O)C5CC5c6cn(nn6)C
8	6QMC	J6H	c1cc(ccc1C(CC(=O)O)C2=CC(=O)NC=C2)Cl
9	6QMD	J6N	Cn1c2ccc(cc2nn1)C(CC(=O)O)c3ccc(cc3)Cl
10	6QME	J6Q	Cc1cc(ccc1Cl)C(CC(=O)O)C2ccc3c(c2)nnn3C
11	6QMJ	J6K	Cc1ccc(cc1CN(C)S(=O)(=O)c2cccc2)C(CC(=O)O)c3cc4c(c(c3)OC)n(nn4)C
12	6QMK	J8H	Cc1ccc(cc1CN2CCOc3cccc3S2(=O)=O)C(CC(=O)O)c4cc5c(c(c4)OC)n(nn5)C
13	5FNQ	SOW	c1cc(ccc1CCC(=O)O)Cl
14	5FNR	XMS	Cn1c2ccc(cc2nn1)C(CC(=O)O)c3ccc(cc3)Cl
15	5FNS	XYX	Cn1c2ccc(cc2nn1)C(CC(=O)O)c3ccc(c(c3)CN(C)S(=O)(=O)C)Cl
16	5FNT	OPL	Cc1c2ccc(cc2nn1)C(CC(=O)O)c3ccc(c(c3)CN(C)S(=O)(=O)c4cccc4)Cl
17	5FNU	L6I	Cc1ccc(cc1CN2CC(Oc3cccc3S2(=O)=O)C)C(CC(=O)O)c4cc5c(c(c4)OC)n(nn5)C

18	5FZJ	75K	<chem>Cc1nc2c(o1)C=C(OC2=O)C</chem>
19	5FZN	FB2	<chem>c1ccc(cc1)S(=O)(=O)N</chem>
20	6V6Z	Q5Y	<chem>COc1ccc(cc1)S(=O)(=O)Nc2ccc(c3c2cccc3)N(CC(=O)O)S(=O)(=O)c4ccc(cc4)OC</chem>
21	7XM5	6i	<chem>O=S(N(CC(N)=O)C1=C2C(C=CC=C2)=C(N(CC(N)=O)S(=O)(=O)C3=CC=C(NC(CCN4CCOCC4)=O)C=C3)=O)C=C1)(C5=CC=C(OC)C=C5)=O</chem>
22	7XM3	6k	<chem>O=S(N(CC(N)=O)C1=C2C(C=CC=C2)=C(N(CC(N)=O)S(=O)(=O)C3=CC=C(NC(CCN4CCN(CC)CC4)=O)C=C3)=O)C=C1)(C5=CC=C(OC)C=C5)=O</chem>
23	7XM4	6e	<chem>O=S(N(CC(N)=O)C1=C2C(C=CC=C2)=C(N(CC(N)=O)S(=O)(=O)C3=CC=C(NC(CN4CCN(CC)CC4)=O)C=C3)=O)C=C1)(C5=CC=C(OC)C=C5)=O</chem>
24	7XM2	GFD	<chem>COc1ccc(cc1)S(=O)(=O)N(CC(=O)N)c2ccc(c3c2cccc3)N(CC(=O)N)S(=O)(=O)c4ccc(cc4)N</chem>

Table-4 inhibitors of the RBD-hACE2 PPI that were experimentally identified (Hanson et al. 2020)

Sr. No	Inhibitor Name	SMILES
1	Ciclopirox	<chem>CC=2C=C(C1CCCCC1)N(O)C(=O)C=2.NCCO</chem>
2	CetylpyridiniumBromide	<chem>[Br-].CCCCCCCCCCCCCCCC[n+].1cccc1</chem>
3	CETYL PYRIDINIUM	<chem>CCCCCCCCCCCCCCCC[N+].1=CC=CC=C1</chem>
4	SodiumdeoxycholateMonohydrate	<chem>C[C@H](CCC([O-]))=O[C@H]1CC[C@H]2[C@@H]3CC[C@H]4[C@H](O)CC[C@]4(C)[C@H]3C[C@H](O)[C@]12C</chem>
5	Triethylenetetramine	<chem>NCCNCCNCCN</chem>
6	Hinokitiol	<chem>CC(C)Cv1=C/C=C(/C(=O)\C(=C1)O</chem>
7	Oxfendazole	<chem>O=C(OC)Nc1nc2cc(ccc2n1)S(=O)c3ccccc3</chem>
8	Prulifloxacin	<chem>CC=5OC(=O)OC=5CN1CCN(CC1)c2cc3c(cc2F)C(=O)C(=C4SC(C)N34)C(=O)O</chem>
9	Prulifloxacin	<chem>CC1SC/2=C(C(O)=O)C(=O)C3=CC(=C(C=C3N12)N4CCN(CC4)C\C5=C(C)OC(=O)O5)F</chem>
10	NH125	<chem>CCCCCCCCCCCCCCCC[N+].1=C(C)[N](CC2=CC=CC=C2)C=C1</chem>
11	PicolinicAcid	<chem>O=C(O)c1ccccn1</chem>
12	AgaricAcid	<chem>O=C(O)C(CCCCCCCCCCCCCC)C(O)(CC(=O)O)C(=O)O</chem>
13	Toltrazuril(sulfone)	<chem>CN1C(=O)NC(=O)N(C1=O)C2=CC=C(OC3=CC=C(C=C3)[S+](([O-])(=O)C(F)(F)F)C(=C2)C</chem>
14	9(Z)-HexadecenoicAcid	<chem>CCCCC/C=C/CCCCCCC(O)=O</chem>
15	BenzylhexadecyldimethylammoniumChloride	<chem>[Cl-].[C[N+](C)(Cc1cccc1)CCCCCCCCCCCCCCCC</chem>
16	Corilagin	<chem>OC1C2COC(=O)C3=CC(=C(O)C(=C3C4=C(O)C(=C(O)C=C4C(=O)OC1C(O)C(O2)OC(=O)C5=CC(=C(O)C(=C5)O)O)O)O</chem>
17	Abametapir	<chem>CC1=CN=C(C=C1)C2=NC=C(C)C=C2</chem>
18	Bictegravir	<chem>OC1=C2N(CC3OC4CCC(C4)N3C2=O)C=C(C(=O)NCC5=C(F)C=C(F)C=C5F)C1=O</chem>
19	Mebendazole	<chem>O=C(OC)Nc1nc2cc(ccc2n1)C(=O)c3ccccc3</chem>
20	Fenbendazole	<chem>O=C(OC)Nc1nc2cc(ccc2n1)Sc3ccccc3</chem>
21	EnalaprilMaleate	<chem>OC(=O)[C@@H]2CCCCN2C(=O)[C@H](C)N[C@@H](CCc1cccc1)C(=O)OCC</chem>
22	Entacapone	<chem>Oc1cc(\C=C(/C#N)C(=O)N(CC)CC)cc([N+]([O-])=O)c1O</chem>
23	Dichloro(ethylenediamine)platinum(II)	<chem>[Cl-].[Pt]1([Cl-])NCCN1</chem>
24	ElaidicAcid	<chem>CCCCCCCC/C=C/CCCCCCC(O)=O</chem>
25	Dolutegravir(GSK1349572)	<chem>C[C@H]1CCO[C@H]2CN3/C=C(/C(=O)NCC4=C(F)C=C(F)C=C4)C(=O)\C(=C3C(=O)N12)O</chem>
26	Dolutegravir(GSK1349572)	<chem>C[C@H]1CCO[C@H]2CN3/C=C(/C(=O)NCC4=C(F)C=C(F)C=C4)C(=O)\C(=C3C(=O)N12)O</chem>
27	Cabotegravir	<chem>C[C@H]1CO[C@H]2CN3/C=C(/C(=O)NCC4=C(F)C=C(F)C=C4)C(=O)\C(=C3C(=O)N12)O</chem>
28	Cangrelor(AR-C69931)	<chem>CSCCNC1=NC(=NC2=C1N=C[N]2[C@@H]3O[C@H](CO[P](O)(=O)O[P](O)(=O)C(Cl)(Cl)[P](O)(O)=O)[C@@H](O)[C@H]3O)SCCC(F)(F)F</chem>